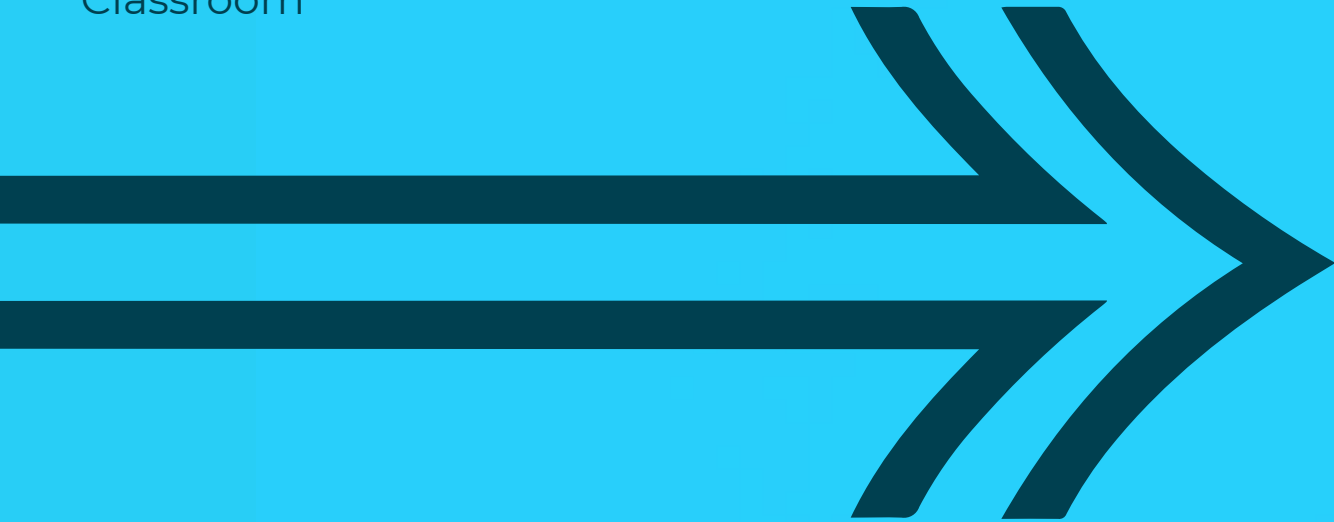




MICROSERVICES AND RESTFUL API

Classroom





MICROSERVICE TRANSACTIONS



Challenges to many microservices

- Coordinate operations
- Handle Failures and Exceptions
- Integrate business logic

Coordination or Orchestration

- To monitor and co-ordinate operations of each microservice collect and hold state information centrally in a datastore e.g. DynamoDB
- Use an orchestration tool to coordinate the microservice integration

AWS Step Functions

- A State Machine that will provide a workflow to orchestrate your business logic



STEP FUNCTION – WHAT CAN I DO?



Is a visual-workflow service to orchestrate services

- Execute more than one Lambda in parallel
- Runs functions in parallel
- Retry functions
- Restart failed Lambdas

Long running code

- Lambda had max timeout 15minutes – Call on external (Aws Services) for long running activities
- Coordinate asynchronous functions
- Manage try-catch-finally
- Sequence functions

Choose function flow based on values



LAB



AWS Step Functions

- Enables you to coordinate the components of distributed applications and microservices using visual workflows.
- Manages the underlying infrastructure to ensure your application is available at any scale.
- In this lab, you will see what AWS Step Functions are and what types of problems this service is best suited to solve.
- Combines Lambda and Step Functions for a gaming application.

Learning Objectives

1. Understand what AWS Step Functions is and how to use it
2. Components of Step Functions and how to configure them
3. Implement Parallel Steps, when to use them and why
4. Understand when best suited to using Conditional Steps
5. Make a State Machine log to the CloudWatch service
6. Create and test a State Machine



Lab Activity

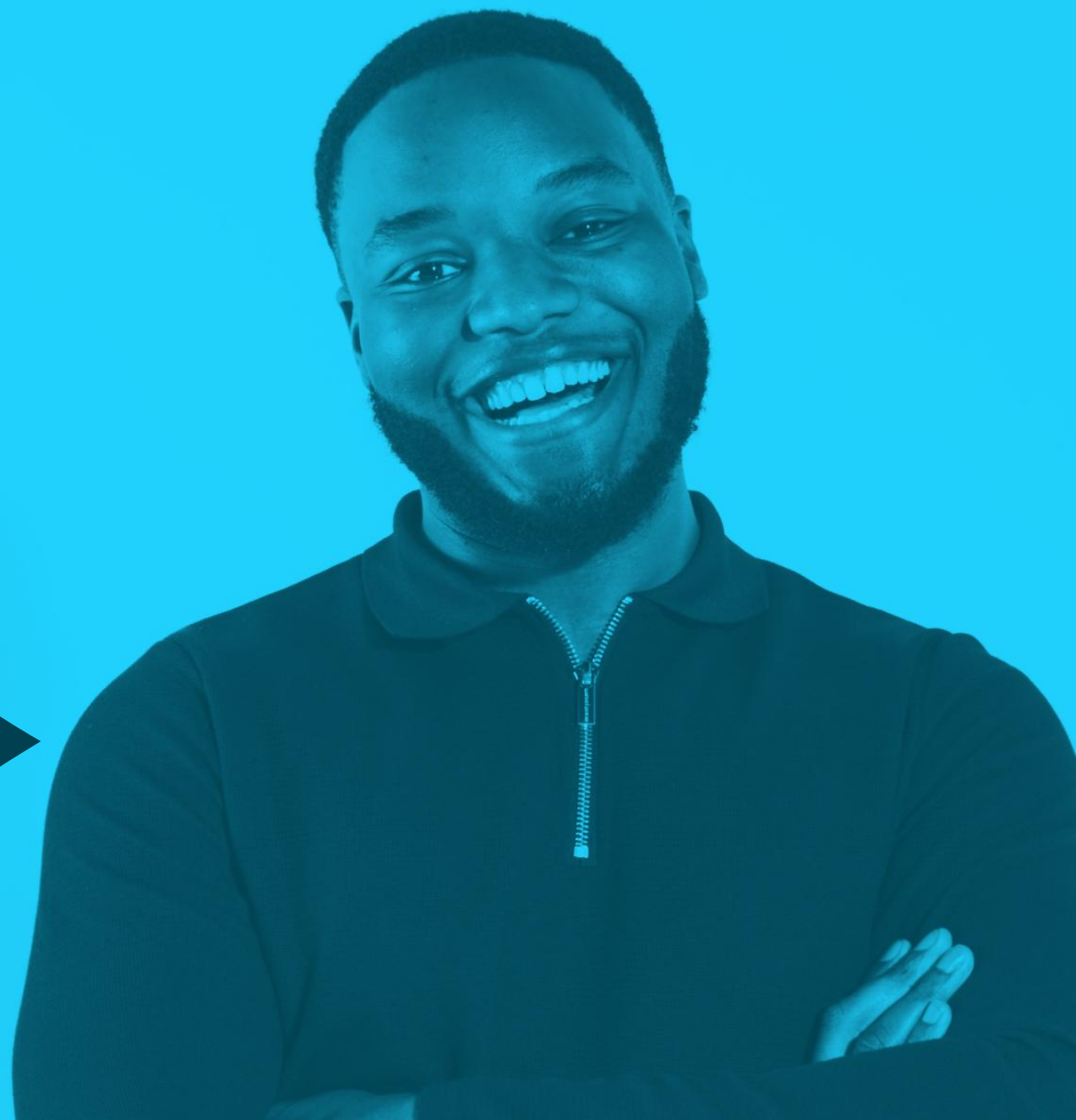
You'll use Cloud Academy to explore using an orchestrator to control the execution of Microservices

The example given uses Lambda, but Step Functions can integrate with other service activities such as Container Services (ECS) etc

Introduction to AWS Step Functions

Duration: 1hr 30 mins







Cart Service: Adds item to cart. Initiates checkout → event for Inventory service



Inventory Service: Reserve item → Event to Cart Service.



Checkout Service: Initiate checkout → Event to Payment Service



Payment Service: Process payment → Event to Order Service.



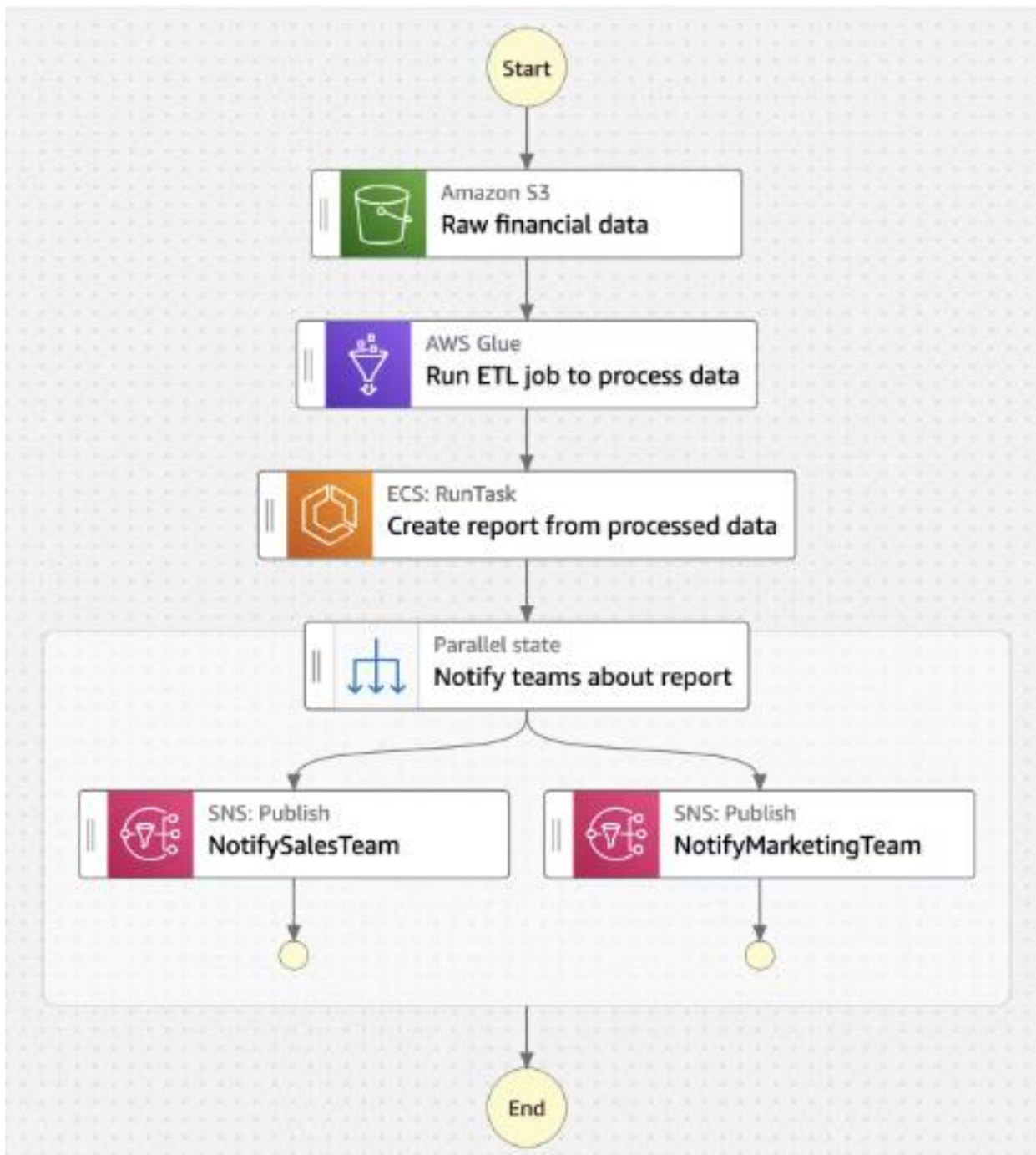
Order Service: Create order → Event to Inventory and Shipping Service.



Shipping Service: Schedule dispatch
→ Event to Notification Service.



Notification Service:
Send user notification





- Drag-and-drop visual editor to assemble microservices into unified workflows.
- Manage input, output, error handling, and retries.
- Developers can focus on business logic rather than writing integration code
- Decouples business logic from the code that defines how your application is implemented.
- Metrics: Activity, Service matrices
 - The number of times a state machine started, succeeded, and failed, and the length of time

- Parallel processing
- Activity task - External may require human approval as well as Service task
- Data can be passed from one task to another.
- Handles failure or success
- Pause time
- Repeat execution for each item of input